# Midterm Submission 2: Satellite Computer Vision for Environmental Impact

Automatic Trophic Early Warning System in the Galapagos Marine Reserve

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## 1 Problem Definition

The El Niño phenomenon causes a substantial temperature change in the ocean currents reaching the Galapagos Marine Reserve (SDG 14: Life Below Water). The increase in Sea Surface Temperature (SST) inhibits the upwelling of cold nutrients, causing the collapse of phytoplankton (chlorophyll-a). This collapse at the base of the food web causes massive starvation in endemic species that rely on this resource. The objective of this project is to develop a Computer Vision system that fuses spatial variables of temperature and chlorophyll to automatically detect these "dead zones" and generate early warning maps.

## 2 Dataset Description

To automate the satellite data collection process, the Google Earth Engine (GEE) Python API was used. The `NASA/OCEANDATA/MODIS-Aqua/L3SMI` collection was queried to extract monthly averages of temperature and Chlorophyll-a exactly within the Galapagos Islands polygon.

In addition, a *Data Cleaning* script was implemented to debug empty or corrupt packets caused by extreme and prolonged cloud cover, resulting in a final validated dataset of 12 temporal pairs (24 images in total) covering recent oceanographic cycles.

## 3 Design

For the design of our Computer Vision model, we have the following key features:

1. **Image Acquisition:** Download of parallel temperature and phytoplankton data via GEE.
2. **Preprocessing:** We used OpenCV's `cv2.resize` to align the tensors, and performed a grayscale conversion to extract the *land* section and isolate only the ocean pixels. A final conversion to the HSV color space was also performed.
3. **Feature Extraction:** Application of deterministic thresholds (`cv2.inRange`) to isolate the positive thermal anomaly (reds/oranges) and severe chlorophyll depletion (deep blue).

4. **Model/Decision Logic:** Multimodal fusion using boolean operations.

# 4 Model Selection and Justification

We chose to use a **Classical Computer Vision** approach based on HSV color thresholds and logical gates (AND), instead of implementing *Deep Learning* architectures like Convolutional Neural Networks.

This decision makes sense because the environmental danger is already indicated by well-defined colors. Using fixed thresholds is much more straightforward, easy to interpret, and does not consume as many computational resources. If we tried to implement Deep Learning with so few images, the model would certainly suffer from *overfitting*, ending up memorizing the data instead of learning to solve the real problem.

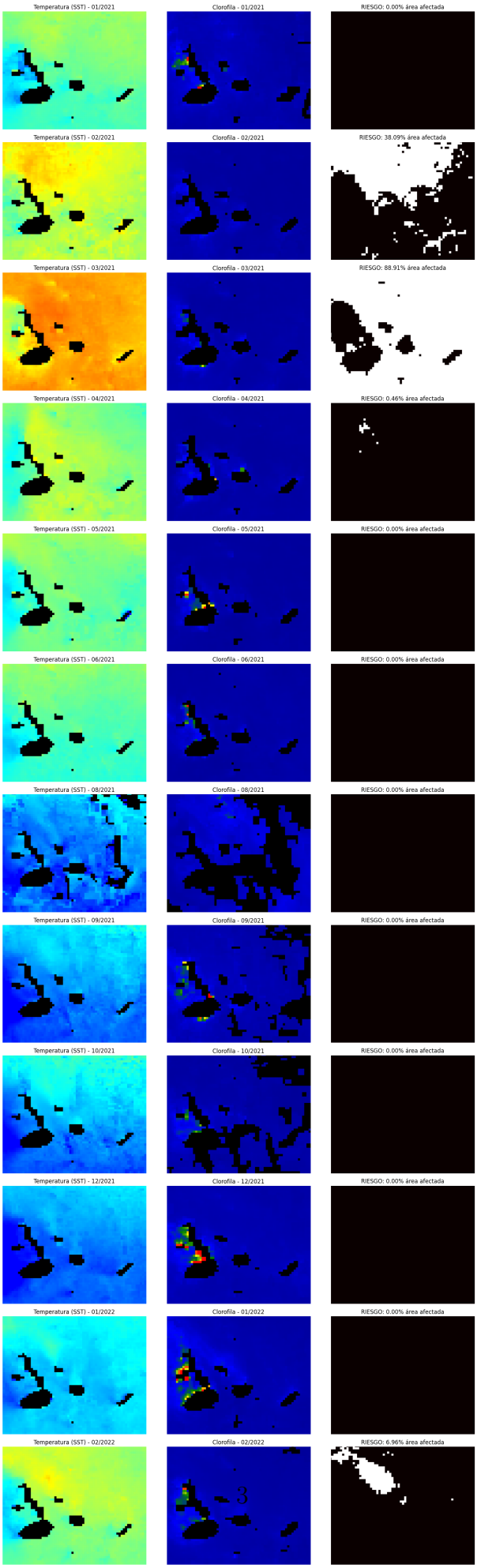
# 5 Results and Visualizations

The model successfully isolates the critical areas. To calculate the impact percentage, we applied this formula:

$$\text{Alert Percentage} = \left( \frac{\text{Pixels in Danger}}{\text{Total Ocean Pixels}} \right) \times 100 \tag{1}$$

The results in Figures 1 and 2 show that the system works quite well. For example, the algorithm autonomously detected a strong marine heatwave in early 2021, where the starvation risk spiked, covering a large part of the reserve, before dropping back to nearly 0% during the colder months.

Resultado Final: Detección de Anomalías Tróficas (SST vs Clorofila)



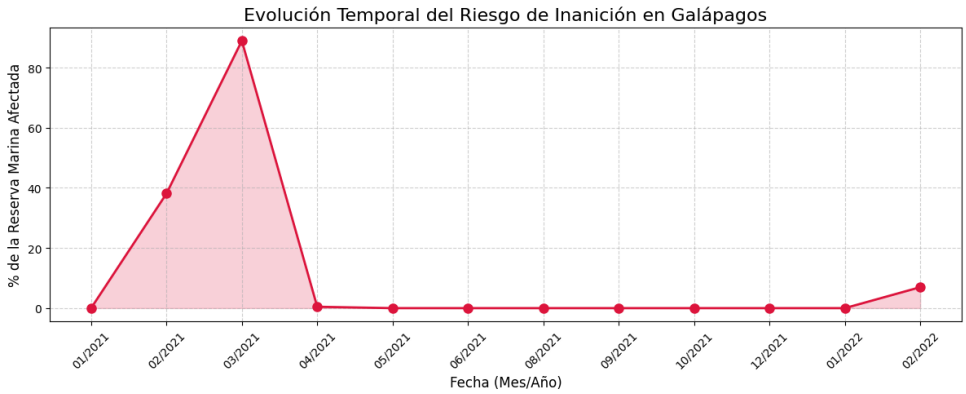



Figure 2: Temporal evolution of the percentage of ocean area at critical risk.

# 6 Limitations and Failures

The main issue or *trade-off* of this method is the cloud cover. Satellites like MODIS cannot see through very dense clouds, so we end up losing data or introducing noise to the image, especially in winter. Another issue is that the model assumes that when the temperature rises, chlorophyll drops immediately, without considering the time it takes for biology to react (*biological lag*). To improve the project in the future, Synthetic Aperture Radar (SAR) data could be integrated, as it is not blocked by the weather.

# 7 Conclusions

In the end, we proved that it is feasible to automate the environmental monitoring of marine areas using classical computer vision techniques and GEE data. The system ends up being a useful, fast, and low-cost tool that could help authorities detect these ecological crises in time.